

# DIAGONAL TRACE IDENTITIES FOR BOSONIC AND FERMIONIC MATRICES

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ABSTRACT. We determine the second stabilised diagonal of finite-rank fermionic trace relations in the  $\partial\psi$  model. In this paper, the formal letters of the model are treated as independent odd matrix letters, denoted  $\psi_0, \psi_1, \psi_2, \dots$ . For every  $N \geq 2$ , the charge- $2N + 2$  relation space at rank  $N$  has odd dimension 4 and even dimension 4. More precisely, it has a basis consisting of the eight signed polarised Cayley–Hamilton trace identities obtained by inserting copies of the even word  $\psi_0^2$ . This proves a conjecture of Eleftheriou, Ji, and Murthy.

## 1. INTRODUCTION

The invariant theory of matrices studies polynomial functions on tuples of matrices which are invariant under simultaneous conjugation. In its classical form, it is a theory of traces and of the relations among traces. For ordinary complex  $N \times N$  matrices, the first fundamental theorem asserts that the invariant ring is generated by traces of words in the matrices, while the second fundamental theorem describes the ideal of relations among these trace generators. In any fixed degree, the cyclic trace words are independent once the rank is sufficiently large. At finite rank this stable independence fails, and the failure is measured by Cayley–Hamilton type trace identities. This point of view goes back to the work of Sibirskii, Razmyslov and Procesi, with important extensions and systematic accounts in Donkin’s work, in Formanek’s monograph, and in the book of De Concini and Procesi [26, 23, 19, 10, 13, 9].

The same finite-rank phenomenon occurs naturally in gauge theory. In free adjoint matrix models, gauge-invariant states are represented by invariant functions of matrices, and partition functions or supersymmetric indices are computed by unitary matrix integrals [27, 1, 17, 25, 15, 18]. Finite- $N$  trace relations are therefore not merely algebraic redundancies: they impose finite-rank constraints on the physical Hilbert space. Recent work has further related such relations to evanescent states, giant-graviton expansions, and analytic continuation in  $N$  [14, 16, 8]. When fermions are present, the relation between trace identities and counting becomes subtler. An index counts bosonic and fermionic states with opposite signs, so a fermionic trace relation may increase, rather than decrease, the value of the index as the rank is lowered.

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Fermionic matrices introduce genuinely new algebraic features. Their entries are Grassmann-valued, the ordinary trace satisfies signed cyclicity rather than ordinary cyclicity, and trace factors supercommute. Thus one must keep track not only of the cyclic order inside each trace, but also of the Koszul signs produced when odd trace factors are interchanged. Already for a single fermionic matrix, signed cyclicity forces traces of even powers to vanish, although the odd matrix itself is not assumed to be square-zero. The resulting trace identities are closely related to the Amitsur–Levitzki identity and to graded trace identity theory [3, 24, 4, 5, 6, 7, 21]. For mixed bosonic and fermionic matrices, one is therefore led to a signed Cayley–Hamilton theory; the signs, however, are not obtained by a formal transcription of the ordinary bosonic formula.

This problem was brought into focus by Eleftheriou, Ji and Murthy [12]. They studied invariants of bosonic and fermionic Grassmann-valued matrices under the adjoint action of  $U(N)$ , with a fermion-number weighting motivated by finite-rank supersymmetric indices. A central example in their work is the  $\partial\psi$  model. We shall write

$$\psi_i := \partial^i \psi, \quad i \geq 0.$$

Thus  $\psi_0 = \psi$ ,  $\psi_1 = \partial\psi$ ,  $\psi_2 = \partial^2\psi$ , and so on. The symbol  $\partial$  is only a label inherited from the physical notation of the model; no derivation is acting on the trace algebra and no Leibniz rule is imposed. Algebraically, the letters

$$\psi_0, \psi_1, \psi_2, \dots$$

are independent odd non-commuting letters with

$$|\psi_i| = 1, \quad \mathbf{q}(\psi_i) = i + 1.$$

We shall distinguish this subscript notation from word powers. For example,  $\psi_0^2 = \psi_0\psi_0$  is a word power, whereas  $\psi_2$  is the independent letter  $\partial^2\psi$ . In index language, the corresponding single-letter index is

$$i_{\partial\psi}(q) = -\frac{q}{1-q} = -q - q^2 - q^3 - \dots$$

This model arises as a simple  $\frac{1}{4}$ -BPS limit of the  $\mathcal{N} = 4$  super Yang–Mills index. A striking feature is that the full index is independent of  $N$ . Since the underlying bosonic and fermionic trace relations do vary with the rank, this  $N$ -independence points to a more refined algebraic question: how are the finite-rank fermionic trace relations organised so that the bosonic and fermionic cancellations occur?

Eleftheriou, Ji and Murthy proposed two structural conjectures for this organisation. The first is a signed polarised Cayley–Hamilton identity for mixed bosonic and fermionic matrices. The second is the corresponding second fundamental theorem, asserting that all trace relations are generated by these signed polarised identities [12, Conjectures 4.1 and 4.2]. Both conjectures were subsequently proved by Procesi [22, Theorems 5.4 and 5.12]. In the same work, Procesi also established the corresponding first fundamental theorem [22, Theorem 4.3], using a graded analogue of the classical symbolic method. Thus the general source of fermionic trace relations is now known.

What remains, and what is needed for the finite- $N$  counting problem, is the more delicate question of minimality in a fixed homogeneous charge. A second fundamental theorem gives generators for the whole  $T$ -ideal of trace identities; it does not, by itself, identify an independent basis for the relation space at a prescribed rank and charge. The present paper settles the first non-basic case of this kind in the  $\partial\psi$  model: the second stabilised diagonal of Pattern I observed in [12]. Along this diagonal the rank and charge are related by

$$(N, n) = (N, 2N + 2), \quad N \geq 2.$$

The computations in [12] predict that the corresponding relation space has dimension  $4 + 4$ , with four odd and four even relations. In the notation above, the predicted relations are obtained by inserting copies of the even word  $\psi_0^2$  into the signed polarised Cayley–Hamilton data. We prove this prediction uniformly in  $N$ .

Here and below, we denote by  $\mathcal{K}_{N,n}$  the charge- $n$  trace-relation space at rank  $N$ , namely the homogeneous charge- $n$  component of the kernel of the evaluation map from the free signed trace algebra of the  $\partial\psi$  model to the rank- $N$  signed trace algebra. Using the notation

$$\Theta_N(A_0; A_1, \dots, A_N)$$

for the signed polarised Cayley–Hamilton trace identity recalled in Section 2, our main result is the following.

**Theorem 1.** *For every  $N \geq 2$ , the following eight elements form a basis of the charge  $2N + 2$  trace-relation space  $\mathcal{K}_{N,2N+2}$  in the  $\partial\psi$  model:*

$$\begin{aligned} \text{odd: } & \Theta_N(\psi_0; (\psi_0^2)^{N-2}, \psi_1, \psi_2), \\ & \Theta_N(\psi_0; (\psi_0^2)^{N-2}, \psi_1, \psi_0^3), \\ & \Theta_N(\psi_0; (\psi_0^2)^{N-1}, \psi_0\psi_1), \\ & \Theta_N(\psi_0; (\psi_0^2)^{N-1}, \psi_1\psi_0), \\ \text{even: } & \Theta_N(\psi_0; (\psi_0^2)^{N-2}, \psi_1, \psi_1\psi_0), \\ & \Theta_N(\psi_0; (\psi_0^2)^{N-2}, \psi_1, \psi_0\psi_1), \\ & \Theta_N(\psi_0; (\psi_0^2)^{N-1}, \psi_2), \\ & \Theta_N(\psi_0; (\psi_0^2)^{N-1}, \psi_0^3). \end{aligned}$$

Here  $(\psi_0^2)^m$  denotes  $m$  consecutive copies of the even word  $\psi_0^2 = \psi_0\psi_0$  in the argument list, and  $(\psi_0^2)^0$  denotes the empty list. In particular,  $\psi_0^2$  is a word power, whereas  $\psi_2$  is the independent letter  $\partial^2\psi$ . Consequently

$$\dim \mathcal{K}_{N,2N+2}^{\bar{1}} = 4, \quad \dim \mathcal{K}_{N,2N+2}^{\bar{0}} = 4.$$

The paper is organised as follows. In Section 2 we recall Grassmann-valued matrices, the free signed trace algebra, the signed polarised Cayley–Hamilton identity, and the eight predicted relations in the  $\partial\psi$  model. We also include several small examples to fix the notation and to illustrate the sign conventions. In Section 3 we prove the spanning

statement by a charge classification, including the necessary products of lower-charge identities. In Section 4 we prove linear independence by coefficient extraction in the free signed trace algebra. The final section completes the proof of the main theorem.

## 2. PRELIMINARIES, NOTATION, AND THE EIGHT RELATIONS

Throughout the paper the ground field is  $\mathbb{C}$ , and all vector spaces, algebras and tensor products are over  $\mathbb{C}$ . We work in characteristic zero.

Let

$$\Lambda = \bigwedge_{\mathbb{C}}(\eta_1, \eta_2, \eta_3, \dots) = \Lambda_{\bar{0}} \oplus \Lambda_{\bar{1}}$$

be the exterior algebra on countably many odd generators. Thus  $\Lambda$  is supercommutative: for homogeneous  $a, b \in \Lambda$ ,

$$ab = (-1)^{|a||b|}ba, \quad |a|, |b| \in \{0, 1\}.$$

The countably generated exterior algebra is only a convenient universal ambient algebra. Every fixed-charge calculation involves only finitely many Grassmann generators, and equivalently one may interpret all identities functorially over arbitrary finite-dimensional exterior algebras.

The matrix algebra  $M_N(\Lambda)$  is  $\mathbb{Z}/2\mathbb{Z}$ -graded by

$$M_N(\Lambda)_{\bar{0}} = M_N(\Lambda_{\bar{0}}), \quad M_N(\Lambda)_{\bar{1}} = M_N(\Lambda_{\bar{1}}).$$

A homogeneous matrix in  $M_N(\Lambda)_{\bar{0}}$  is called **bosonic**, and a homogeneous matrix in  $M_N(\Lambda)_{\bar{1}}$  is called **fermionic**. The trace used in this paper is always the ordinary matrix trace

$$\mathrm{tr}(A) = \sum_{i=1}^N A_{ii},$$

not a supertrace.

If  $A, B \in M_N(\Lambda)$  are homogeneous, then the ordinary trace satisfies the signed cyclic rule

$$(2.1) \quad \mathrm{tr}(AB) = (-1)^{|A||B|} \mathrm{tr}(BA).$$

Indeed,

$$\mathrm{tr}(AB) = \sum_{i,j} A_{ij} B_{ji} = (-1)^{|A||B|} \sum_{i,j} B_{ji} A_{ij} = (-1)^{|A||B|} \mathrm{tr}(BA).$$

More generally, if  $U, V$  are homogeneous words in homogeneous matrices, then

$$\mathrm{tr}(UV) = (-1)^{|U||V|} \mathrm{tr}(VU).$$

Trace values themselves lie in the supercommutative algebra  $\Lambda$ , and therefore

$$(2.2) \quad \mathrm{tr}(U) \mathrm{tr}(V) = (-1)^{|U||V|} \mathrm{tr}(V) \mathrm{tr}(U).$$

**Example 2.1.** The word “fermionic” does not mean that an odd matrix is square-zero. Nor do we impose word relations such as  $AB = -BA$  between odd matrix variables. The signs used in this paper come from the Grassmann parity of the entries, from signed cyclicity of the trace, from supercommutation of trace values, and from the graded symmetry of the signed polarised Cayley–Hamilton identity.

Let  $\eta_1, \eta_2$  be odd Grassmann generators and consider the odd matrix

$$A = \begin{pmatrix} 0 & \eta_1 \\ \eta_2 & 0 \end{pmatrix} \in M_2(\Lambda)_{\bar{1}}.$$

Then

$$A^2 = \begin{pmatrix} \eta_1\eta_2 & 0 \\ 0 & \eta_2\eta_1 \end{pmatrix} = \begin{pmatrix} \eta_1\eta_2 & 0 \\ 0 & -\eta_1\eta_2 \end{pmatrix},$$

which is generally non-zero. However,

$$\text{tr}(A^2) = \eta_1\eta_2 + \eta_2\eta_1 = 0.$$

Thus the vanishing of traces such as  $\text{tr}(W^{2a})$  for an odd word  $W$  is a signed-cyclicity phenomenon, not a consequence of an assumption  $W^2 = 0$ .

We now pass to the universal signed trace algebra. Let  $\mathcal{A}$  be an alphabet of homogeneous non-commuting letters. A word is a finite product of letters from  $\mathcal{A}$ . The empty word is denoted by  $\mathbf{1}$ . The parity of a homogeneous word  $W = a_1 \cdots a_r$  is

$$|W| = |a_1| + \cdots + |a_r| \pmod{2}, \quad |\mathbf{1}| = 0.$$

When  $\mathcal{A}$  is also charged, the charge is extended additively to words:

$$q(a_1 \cdots a_r) = q(a_1) + \cdots + q(a_r), \quad q(\mathbf{1}) = 0.$$

The **free signed trace algebra** on  $\mathcal{A}$ , denoted by  $\mathcal{T}(\mathcal{A})$ , is the supercommutative  $\mathbb{C}$ -algebra generated by formal symbols

$$\text{tr}(W), \quad W \neq \mathbf{1},$$

where  $\text{tr}(W)$  is declared to be homogeneous of parity  $|W|$ , modulo the signed cyclic relations

$$(2.3) \quad \text{tr}(UV) = (-1)^{|U||V|} \text{tr}(VU)$$

for homogeneous words  $U, V$ . If the alphabet is charged, we also declare

$$q \text{tr}(W) = q(W),$$

and extend charge additively to products of trace factors. Multiplication in  $\mathcal{T}(\mathcal{A})$  is supercommutative:

$$(2.4) \quad \text{tr}(U) \text{tr}(V) = (-1)^{|U||V|} \text{tr}(V) \text{tr}(U).$$

Thus the square of an odd trace factor is zero. This statement concerns products of trace values in the supercommutative algebra; it should not be confused with a square-zero relation for odd words or odd matrices.

The following elementary consequence of signed cyclicity will be used repeatedly. If  $W$  is an odd word, then

$$(2.5) \quad \mathrm{tr}(W^{2a}) = 0 \quad (a \geq 1).$$

Indeed,

$$\mathrm{tr}(W^{2a}) = \mathrm{tr}(W W^{2a-1}) = (-1)^{1 \cdot (2a-1)} \mathrm{tr}(W^{2a-1} W) = -\mathrm{tr}(W^{2a}).$$

A rank- $N$  homogeneous evaluation of  $\mathcal{A}$  is a parity-preserving assignment

$$\rho : \mathcal{A} \longrightarrow M_N(\Lambda),$$

extended multiplicatively to words. It induces an evaluation homomorphism

$$\mathrm{ev}_{\rho, N} : \mathcal{T}(\mathcal{A}) \longrightarrow \Lambda, \quad \mathrm{ev}_{\rho, N}(\mathrm{tr}(W)) = \mathrm{tr}(\rho(W)).$$

A **signed trace relation** at rank  $N$  is an element of  $\mathcal{T}(\mathcal{A})$  which vanishes under every rank- $N$  homogeneous evaluation. We write

$$\mathcal{K}_N(\mathcal{A}) = \bigcap_{\rho} \ker(\mathrm{ev}_{\rho, N})$$

for the ideal of all rank- $N$  signed trace relations, and

$$\mathcal{T}_N(\mathcal{A}) = \mathcal{T}(\mathcal{A}) / \mathcal{K}_N(\mathcal{A})$$

for the corresponding rank- $N$  signed trace algebra.

If  $\mathcal{A}$  is charged, then  $\mathcal{T}(\mathcal{A})$  is graded by charge. The kernel is homogeneous for this grading: applying an evaluation after replacing each letter  $a$  by  $t^{q(a)}\rho(a)$  separates the coefficients of different powers of the auxiliary parameter  $t$ . We denote the homogeneous charge- $n$  component by

$$\mathcal{K}_{N,n}(\mathcal{A}) = \mathcal{K}_N(\mathcal{A}) \cap \mathcal{T}(\mathcal{A})_n.$$

Its even and odd parts are denoted by  $\mathcal{K}_{N,n}^0(\mathcal{A})$  and  $\mathcal{K}_{N,n}^1(\mathcal{A})$ .

We next recall the signed polarised Cayley–Hamilton expression. In the definition below,  $A_1, \dots, A_N$  are homogeneous words, not necessarily distinct. For a permutation  $\pi \in S_N$ , let

$$\varepsilon(\pi; A_1, \dots, A_N)$$

be the **Koszul sign** obtained by permuting the ordered list  $(A_1, \dots, A_N)$  into  $(A_{\pi(1)}, \dots, A_{\pi(N)})$ . Equivalently,  $\varepsilon(\pi; A_1, \dots, A_N) = (-1)^m$ , where  $m$  is the number of inversions of  $\pi$  involving two odd entries.

If  $\mu = (\mu_1, \dots, \mu_h)$  is a partition of  $r$ , write

$$z_\mu = \prod_{j \geq 1} j^{m_j} m_j!,$$

where  $m_j$  is the multiplicity of the part  $j$  in  $\mu$ , and write  $\ell(\mu) = h$ .

**Definition 2.2.** The signed polarised **Cayley–Hamilton polynomial** of order  $N$  is

$$\begin{aligned} \text{PCH}_N(A_1, \dots, A_N) &= \sum_{r=0}^N \sum_{\mu \vdash r} \frac{(-1)^{\ell(\mu)}}{z_\mu} \frac{1}{N!} \sum_{\pi \in S_N} \varepsilon(\pi; A_1, \dots, A_N) \\ &\quad \times \text{tr}(A_{\pi(1)} \cdots A_{\pi(\mu_1)}) \text{tr}(A_{\pi(\mu_1+1)} \cdots A_{\pi(\mu_1+\mu_2)}) \cdots \\ (2.6) \quad &\quad \times \text{tr}(A_{\pi(r-\mu_h+1)} \cdots A_{\pi(r)}) A_{\pi(r+1)} \cdots A_{\pi(N)}. \end{aligned}$$

For  $r = 0$ , the product of trace factors is interpreted as 1. For  $r = N$ , the remaining non-trace word is interpreted as the empty word **1**. We put

$$(2.7) \quad \Theta_N(A_0; A_1, \dots, A_N) = \text{tr}(A_0 \text{PCH}_N(A_1, \dots, A_N)).$$

Thus  $\text{PCH}_N(A_1, \dots, A_N)$  is a trace-valued non-commutative polynomial, whereas  $\Theta_N(A_0; A_1, \dots, A_N)$  is a trace polynomial. In expanding  $\Theta_N$ , we move the trace factors appearing in (2.6) to the left of the final trace in (2.7). If the product of those trace factors has parity  $s$ , moving it past  $A_0$  contributes the additional sign  $(-1)^{s|A_0|}$ .

Although the notation  $\Theta_N(A_0; A_1, \dots, A_N)$  singles out an outer slot, the expression is a trace form of the signed fundamental trace identity in  $N + 1$  homogeneous slots. We record this explicitly.

For homogeneous  $B_0, \dots, B_N$ , and for  $\sigma \in S_{\{0,1,\dots,N\}}$ , write every cycle of  $\sigma$  starting with its smallest element, and order the cycles by these smallest elements. For a cycle  $c = (i_1 \cdots i_k)$ , put

$$\text{tr}_c(B) = \text{tr}(B_{i_1} \cdots B_{i_k}).$$

Let  $\text{Tr}_\sigma(B_0, \dots, B_N)$  be the product of the trace factors  $\text{tr}_c(B)$ , in the canonical cycle order. Finally, let  $\chi(\sigma; B_0, \dots, B_N)$  be the ordinary sign of  $\sigma$  multiplied by the Koszul sign obtained by reading the entries  $B_i$  in the canonical parenthesised cycle order.

Define

$$(2.8) \quad \Phi_{N+1}(B_0, \dots, B_N) = \sum_{\sigma \in S_{\{0,\dots,N\}}} \chi(\sigma; B_0, \dots, B_N) \text{Tr}_\sigma(B_0, \dots, B_N).$$

**Proposition 2.3.** For homogeneous  $A_0, \dots, A_N$ ,

$$(2.9) \quad \Theta_N(A_0; A_1, \dots, A_N) = \frac{(-1)^N}{N!} \Phi_{N+1}(A_0, A_1, \dots, A_N).$$

In particular,  $\Theta_N$  is graded symmetric in its  $N + 1$  slots.

*Proof.* Separate in the cycle sum (2.8) the cycle which contains the index 0. Rotate this cycle so that 0 is first, and regard the remaining entries in that cycle as the non-trace tail of  $\text{PCH}_N$ . The other cycles give the trace factors. Grouping the remaining cycles by their cycle type  $\mu$  gives the factors  $z_\mu^{-1}$ , and the average over the  $N$  inner slots gives the factor  $1/N!$ . The ordinary cycle signs give the global scalar  $(-1)^N$  relative to (2.6). The Koszul signs are exactly the signs  $\varepsilon(\pi; A_1, \dots, A_N)$ , together with the signs needed to move the trace factors to the left of the final trace. This proves (2.9).

The cycle sum is manifestly graded symmetric: permuting two homogeneous slots changes the expression by the corresponding Koszul sign and relabels the summation over  $S_{\{0, \dots, N\}}$ .  $\square$

**Lemma 2.4.** *If two slots of  $\Theta_N(A_0; A_1, \dots, A_N)$  are equal odd words, then*

$$\Theta_N(A_0; A_1, \dots, A_N) = 0.$$

*Proof.* Interchanging the two equal odd slots leaves the displayed expression unchanged as a formal trace polynomial, but graded symmetry multiplies it by  $-1$ . Since the ground field has characteristic zero, the expression vanishes.  $\square$

**Lemma 2.5.** *If one of the  $N + 1$  slots of  $\Theta_N(A_0; A_1, \dots, A_N)$  is the identity word  $\mathbf{1}$ , then the identity is zero after imposing  $\text{tr}(\mathbf{1}) = N$ .*

*Proof.* By graded symmetry we may assume that the identity word is  $A_0$ . In the cycle form, terms in which 0 is a singleton cycle give a factor  $\text{tr}(\mathbf{1}) = N$ . For each fixed permutation of the remaining  $N$  labels, there are exactly  $N$  ways to insert 0 into one of its cycles. Since  $\mathbf{1}$  is even, these insertions do not change the trace monomial or create Koszul signs, but they reverse the ordinary permutation sign. Thus the  $N$  non-singleton insertions cancel the singleton contribution. Hence the whole expression is zero.  $\square$

We shall use Procesi's signed second fundamental theorem in the following form. A  $T$ -ideal of signed trace polynomials means an ideal stable under homogeneous substitutions of words for letters and under multiplication by arbitrary signed trace polynomials.

**Theorem 2** ([22, Theorem 5.12]). *For  $N \times N$  bosonic and fermionic Grassmann-valued matrices, the ideal of signed trace relations is generated, as a  $T$ -ideal, by the trace identities*

$$\Theta_N(A_0; A_1, \dots, A_N)$$

*with homogeneous slots.*

We now specialise to the model used in the rest of the paper. Let

$$\mathcal{A}_{\partial\psi} = \{\psi_i \mid i \geq 0\}, \quad \psi_i := \partial^i \psi.$$

Every letter in  $\mathcal{A}_{\partial\psi}$  is fermionic, and its charge is

$$q(\psi_i) = i + 1.$$

The notation  $\partial^i \psi$  is retained only to indicate the origin of the model. Algebraically,  $\partial$  is not an operator on the trace algebra. No derivation and no Leibniz rule are imposed. Thus

$$\psi_0, \psi_1, \psi_2, \dots$$

are independent odd non-commuting letters with prescribed charges.



We shall carefully distinguish subscripts from word powers. The subscript in  $\psi_i$  labels the independent letter  $\partial^i\psi$ . Superscripts denote products in the free word algebra; for example,

$$\psi_0^2 = \psi_0\psi_0, \quad \psi_0^3 = \psi_0\psi_0\psi_0,$$

whereas  $\psi_2$  is the single letter  $\partial^2\psi$ .

From now on we abbreviate

$$\mathcal{T} := \mathcal{T}(\mathcal{A}_{\partial\psi}), \quad \mathcal{K}_{N,n} := \mathcal{K}_{N,n}(\mathcal{A}_{\partial\psi}).$$

Thus  $\mathcal{K}_{N,n}$  is the charge- $n$  part of the kernel of all rank- $N$  evaluations of the free signed trace algebra of the  $\partial\psi$  model. Its even and odd parts are denoted by  $\mathcal{K}_{N,n}^0$  and  $\mathcal{K}_{N,n}^1$  respectively.

The non-empty words of charge at most 3 are exactly

$$\psi_0, \quad \psi_0^2, \psi_1, \quad \psi_2, \psi_0^3, \psi_0\psi_1, \psi_1\psi_0.$$

Their parities are

$$\psi_0, \psi_1, \psi_2, \psi_0^3 \quad \text{odd},$$

and

$$\psi_0^2, \psi_0\psi_1, \psi_1\psi_0 \quad \text{even}.$$

For  $N \geq 2$ , define four odd trace relations

$$(2.10) \quad F_1 = \Theta_N(\psi_0; (\psi_0^2)^{N-2}, \psi_1, \psi_2), \quad F_2 = \Theta_N(\psi_0; (\psi_0^2)^{N-2}, \psi_1, \psi_0^3),$$

$$(2.11) \quad F_3 = \Theta_N(\psi_0; (\psi_0^2)^{N-1}, \psi_0\psi_1), \quad F_4 = \Theta_N(\psi_0; (\psi_0^2)^{N-1}, \psi_1\psi_0),$$

and four even trace relations

$$(2.12) \quad G_1 = \Theta_N(\psi_0; (\psi_0^2)^{N-2}, \psi_1, \psi_1\psi_0), \quad G_2 = \Theta_N(\psi_0; (\psi_0^2)^{N-2}, \psi_1, \psi_0\psi_1),$$

$$(2.13) \quad G_3 = \Theta_N(\psi_0; (\psi_0^2)^{N-1}, \psi_2), \quad G_4 = \Theta_N(\psi_0; (\psi_0^2)^{N-1}, \psi_0^3).$$

Here  $(\psi_0^2)^m$  means  $m$  consecutive slots equal to the even word  $\psi_0^2$  in the argument list of  $\Theta_N$ ; when  $m = 0$ , the list is empty. Each of the eight relations has total charge  $2N + 2$ . Their stated parities are the total parities of the  $N + 1$  slots.

**Remark 2.6.** The letter  $\psi_3 = \partial^3\psi$  has charge 4. It does not occur in the eight relations above, and this is not because of a hidden elimination formula. Rather, on the charge- $2N + 2$  diagonal, every primitive candidate containing  $\psi_3$  is already zero.

Indeed, a primitive identity

$$\Theta_N(W_0; W_1, \dots, W_N)$$

has  $N + 1$  non-empty slots. The baseline charge on this diagonal is 2 per slot, giving total charge  $2N + 2$ . A slot equal to  $\psi_3$  has charge 4, hence excess +2 relative to this baseline. To keep total charge  $2N + 2$ , the remaining slots would have to include two charge-one slots. The only charge-one word is  $\psi_0$ , and it is odd. Therefore such a candidate has two equal odd slots  $\psi_0, \psi_0$ , so it vanishes by graded symmetry.

For example,

$$\Theta_N(\psi_0; \psi_0, (\psi_0^2)^{N-2}, \psi_3) = 0.$$

The same argument applies a fortiori if a slot contains  $\psi_3$  together with other letters. Such a slot has charge at least 5; hence either the charge equation has no solution, or the required compensation forces at least two charge-one slots, both equal to the odd word  $\psi_0$ . The corresponding identity then vanishes by graded symmetry.

Trace-multiple contributions cannot introduce  $\psi_3$  either. By the lower-charge classification proved below, the only non-zero primitive identities below charge  $2N + 2$  have charge  $2N + 1$ , so the multiplier would have to have charge 1 and hence cannot contain  $\psi_3$ .

Thus no non-zero relation involving  $\psi_3$  has to be eliminated on this diagonal.

**Remark 2.7.** The letter  $\psi_2$  has charge 3 and may occur as a single charge-3 slot. This is why it appears in

$$F_1 = \Theta_N(\psi_0; (\psi_0^2)^{N-2}, \psi_1, \psi_2)$$

and

$$G_3 = \Theta_N(\psi_0; (\psi_0^2)^{N-1}, \psi_2).$$

However, two separate slots equal to  $\psi_2$  would be two equal odd slots, and the corresponding identity would vanish by graded symmetry.

Word powers of  $\psi_2$  do not give new primitive relations on the charge- $2N + 2$  diagonal. For instance, the word  $\psi_2^2$  has charge 6, which has excess +4 relative to the baseline charge 2 for a slot. If the remaining slots cannot compensate this excess, no such candidate exists. If they can, then at least two charge-one slots must occur; both are equal to the odd word  $\psi_0$ , and the candidate vanishes by graded symmetry. Thus the relevant occurrence of  $\psi_2$  on this diagonal is the single charge-3 slot displayed above.

These eight signed polarised Cayley–Hamilton trace identities are precisely the relations predicted by Eleftheriou, Ji and Murthy in the second stabilised diagonal of Pattern I [12]. The problem is not merely to write down these identities, but to prove that they give the whole homogeneous relation space and that no linear dependence is hidden among them. In the notation fixed above, the goal is to prove

$$\mathcal{K}_{N,2N+2} = \text{span}\{F_1, F_2, F_3, F_4, G_1, G_2, G_3, G_4\},$$

with the eight displayed generators linearly independent.

The first point of the diagonal is  $N = 2$ . In that case  $(\psi_0^2)^{N-2}$  is the empty list. Explicitly, the eight relations are

$$\begin{aligned} F_1 &= \Theta_2(\psi_0; \psi_1, \psi_2), & F_2 &= \Theta_2(\psi_0; \psi_1, \psi_0^3), \\ F_3 &= \Theta_2(\psi_0; \psi_0^2, \psi_0\psi_1), & F_4 &= \Theta_2(\psi_0; \psi_0^2, \psi_1\psi_0), \\ G_1 &= \Theta_2(\psi_0; \psi_1, \psi_1\psi_0), & G_2 &= \Theta_2(\psi_0; \psi_1, \psi_0\psi_1), \\ G_3 &= \Theta_2(\psi_0; \psi_0^2, \psi_2), & G_4 &= \Theta_2(\psi_0; \psi_0^2, \psi_0^3). \end{aligned}$$

Passing from rank  $N$  to rank  $N + 1$  inserts one further copy of the even word  $\psi_0^2$  into the Cayley–Hamilton data and raises the charge by 2.

**Two guiding examples.** We record two elementary examples before entering the general spanning argument. They explain the simplifications used repeatedly below.

First, put

$$X = \psi_0^2.$$

The word  $X$  is not zero. However, every trace block containing only powers of  $X$  vanishes:

$$\mathrm{tr}(X^a) = \mathrm{tr}(\psi_0^{2a}) = 0, \quad a \geq 1.$$

Indeed,  $\psi_0$  is odd, so signed cyclicity gives

$$\mathrm{tr}(\psi_0^{2a}) = \mathrm{tr}(\psi_0 \psi_0^{2a-1}) = -\mathrm{tr}(\psi_0^{2a-1} \psi_0) = -\mathrm{tr}(\psi_0^{2a}).$$

Thus  $\mathrm{tr}(\psi_0^{2a}) = 0$  in characteristic zero.

Consequently, in

$$\Theta_N(\psi_1; X^N)$$

all proper trace-block terms vanish, because every such block contains only powers of  $X$ . Only the tail term survives, and hence

$$\Theta_N(\psi_1; X^N) = \mathrm{tr}(\psi_0^{2N} \psi_1).$$

On the other hand, the explicit expansions of  $F_3$  and  $F_4$  give

$$F_3 + F_4 = 2 \mathrm{tr}(\psi_0^{2N} \psi_1).$$

Therefore

$$\Theta_N(\psi_1; (\psi_0^2)^N) = \frac{1}{2}(F_3 + F_4).$$

This is the simplest example of a primitive identity not listed separately among the eight displayed generators but lying in their span.

A second useful example is

$$G_4 = \Theta_N(\psi_0; X^{N-1}, \psi_0^3).$$

In its expansion, every non-zero proper trace-block term has one trace block containing the slot  $\psi_0^3$ . The terms indexed by  $r$  and  $N - r$  give the same product of two odd trace factors in the opposite order, and hence cancel after supercommuting the trace factors. The only unpaired endpoint term remains. Thus

$$G_4 = -\frac{1}{N} \mathrm{tr}(\psi_0) \mathrm{tr}(\psi_0^{2N+1}).$$

This cancellation is typical of the signed trace computations used below.

### 3. SPANNING IN CHARGE $2N + 2$

We first prove that the eight displayed identities span the whole charge- $2N + 2$  relation space.

We shall use the following elementary consequence of the  $T$ -ideal statement throughout this section.

**Lemma 3.1.** *Let  $n \geq 0$ . The charge- $n$  component  $\mathcal{K}_{N,n}$  is spanned by elements of the form*

$$(3.14) \quad M \cdot \Theta_N(W_0; W_1, \dots, W_N),$$

where  $M$  is a trace monomial, the  $W_i$  are non-empty homogeneous words in the  $\partial\psi$  alphabet, and

$$(3.15) \quad \text{q } M + \sum_{i=0}^N \text{q } W_i = n.$$

*Proof.* By the signed second fundamental theorem, the kernel is generated as a  $T$ -ideal by the identities  $\Theta_N(A_0; A_1, \dots, A_N)$  with homogeneous slots. Since all letters in the  $\partial\psi$  alphabet have positive charge, the charge- $n$  part of any  $T$ -ideal combination is obtained by taking homogeneous substitutions and homogeneous trace-monomial multipliers. If a substituted slot is the empty word  $\mathbf{1}$ , then the corresponding identity is zero after imposing  $\text{tr}(\mathbf{1}) = N$ , by Lemma 2.5. Hence the non-zero contributions may be taken to have all slots non-empty. This gives (3.14) and the charge constraint (3.15).  $\square$

A relation of the form  $\Theta_N(W_0; W_1, \dots, W_N)$  will be called **primitive** if each  $W_i$  is a non-empty homogeneous word. Its charge is the sum of the charges of its  $N + 1$  slots.

Before classifying the primitive identities, we record the elementary expansions which will be used both in the spanning argument and in the independence argument. Put, only in this proof,  $X = \psi_0^2$ ,  $D = \psi_1$ , and  $Y = \psi_0^3$ .

**Lemma 3.2.** *For every  $N \geq 2$ , the following identities hold in the free signed trace algebra:*

$$(3.16) \quad \Theta_N(D; X^N) = \text{tr}(\psi_0^{2N} \psi_1),$$

$$(3.17) \quad \Theta_N(\psi_0; X^N) = \text{tr}(\psi_0^{2N+1}),$$

$$(3.18) \quad \Theta_N(\psi_0; X^{N-1}, D) = \text{tr}(\psi_0^{2N-1} \psi_1) - \frac{1}{N} \sum_{a=0}^{N-1} \text{tr}(\psi_0^{2a+1}) \text{tr}(\psi_0^{2N-2a-2} \psi_1),$$

$$(3.19) \quad F_3 = \text{tr}(\psi_0^{2N} \psi_1) - \frac{1}{N} \sum_{a=0}^{N-1} \text{tr}(\psi_0^{2a+1}) \text{tr}(\psi_0^{2N-2a-1} \psi_1),$$

$$(3.20) \quad F_4 = \text{tr}(\psi_0^{2N} \psi_1) + \frac{1}{N} \sum_{a=0}^{N-1} \text{tr}(\psi_0^{2a+1}) \text{tr}(\psi_0^{2N-2a-1} \psi_1),$$

$$(3.21) \quad \begin{aligned} F_2 &= \frac{1}{N} \text{tr}(\psi_0) \text{tr}(\psi_0^{2N-1} \psi_1) - \frac{1}{N(N-1)} \sum_{a=1}^{N-1} \text{tr}(\psi_0^{2a+1}) \text{tr}(\psi_0^{2N-2a-1} \psi_1) \\ &\quad - \frac{1}{N(N-1)} \sum_{a=1}^{N-1} \text{tr}(\psi_0) \text{tr}(\psi_0^{2a+1}) \text{tr}(\psi_0^{2N-2a-2} \psi_1), \end{aligned}$$

$$(3.22) \quad G_4 = -\frac{1}{N} \text{tr}(\psi_0) \text{tr}(\psi_0^{2N+1}).$$

Consequently,

$$(3.23) \quad F_3 + F_4 = 2 \text{tr}(\psi_0^{2N} \psi_1).$$

*Proof.* We expand the defining formula for  $\text{PCH}_N$ . Any trace block which contains only powers of  $X = \psi_0^2$  gives

$$\text{tr}(X^a) = \text{tr}(\psi_0^{2a}) = 0 \quad (a \geq 1),$$

by signed cyclicity. This simple observation eliminates most terms.

For  $\Theta_N(D; X^N)$  and  $\Theta_N(\psi_0; X^N)$ , every proper trace block contains only powers of  $X$ , and hence vanishes. Therefore only the tail term survives, giving (3.16) and (3.17).

We next consider a single distinguished slot among copies of  $X$ . A one-block trace contribution of length  $r$ , containing the distinguished slot and  $r-1$  copies of  $X$ , has coefficient

$$-\frac{1}{r} \cdot \frac{r(N-1)!}{N!} = -\frac{1}{N}$$

before the extra sign coming from moving the trace block past the outer slot. Applying this count and then putting the trace factors in the fixed canonical order gives (3.18), (3.19) and (3.20). The difference between the signs in (3.19) and (3.20) is exactly the signed cyclic relation

$$\text{tr}(\psi_0^{2a} \psi_1 \psi_0) = -\text{tr}(\psi_0^{2a+1} \psi_1).$$

For  $F_2$ , there are two odd defect slots,  $D = \psi_1$  and  $Y = \psi_0^3$ . A non-zero proper term has either one trace block containing both  $D$  and  $Y$ , or two trace blocks, one containing

$D$  and the other containing  $Y$ . The first possibility gives the two-trace terms in (3.21); the all-inner one-block term is the special term

$$\frac{1}{N} \operatorname{tr}(\psi_0) \operatorname{tr}(\psi_0^{2N-1} \psi_1).$$

The second possibility gives the three-trace sum in (3.21). The coefficient  $1/(N(N-1))$  is the labelled polarisation count for distributing the two distinguished odd slots into the relevant trace block pattern, and the displayed signs are the Koszul signs obtained after signed cyclic normalisation.

It remains only to justify (3.22), since here the vanishing is not termwise. In

$$G_4 = \Theta_N(\psi_0; X^{N-1}, Y),$$

every non-zero proper trace-block term has exactly one trace block, and that block must contain  $Y$ . Thus only the one-part partitions  $(r)$ ,  $1 \leq r \leq N$ , contribute. The term with a block containing  $Y$  and  $r-1$  copies of  $X$  is, before canonical ordering,

$$\frac{1}{N} \operatorname{tr}(\psi_0^{2r+1}) \operatorname{tr}(\psi_0^{2N-2r+1}).$$

For  $1 \leq r \leq N-1$ , the terms indexed by  $r$  and  $N-r$  are the same product of two odd trace factors in the opposite order; after supercommuting them into canonical order they cancel. If  $r = N-r$ , the term is the square of an odd trace factor and is zero. The only unpaired term is  $r = N$ , and putting  $\operatorname{tr}(\psi_0^{2N+1}) \operatorname{tr}(\psi_0)$  into canonical order gives the extra minus sign. Hence

$$G_4 = -\frac{1}{N} \operatorname{tr}(\psi_0) \operatorname{tr}(\psi_0^{2N+1}).$$

Finally, (3.23) follows immediately by adding (3.19) and (3.20).  $\square$

**Lemma 3.3.** *Let  $N \geq 2$ . Every non-zero primitive identity of charge  $2N+2$  lies in the span of*

$$F_1, F_2, F_3, F_4, G_1, G_2, G_3, G_4.$$

*Proof.* A primitive identity has  $N+1$  non-empty slots. The unique word of charge 1 is  $\psi_0$ , and it is odd. The words of charge 2 are

$$\psi_0^2, \quad \psi_1,$$

where  $\psi_0^2$  is even and  $\psi_1$  is odd. The words of charge 3 are

$$\psi_2, \quad \psi_0^3, \quad \psi_0 \psi_1, \quad \psi_1 \psi_0.$$

If at least two slots have charge 1, then two slots are equal to the same odd word  $\psi_0$ . The identity therefore vanishes by Lemma 2.4. Suppose next that exactly one slot has charge 1. By graded symmetry we may move this slot to the outer position, up to the Koszul sign. The remaining  $N$  slots have total charge  $2N+1$ , and each has charge at least 2. Their charges must therefore be  $2, \dots, 2, 3$ . Among the charge-2 slots, the odd word  $\psi_1$  can occur at most once; two occurrences would again force vanishing by Lemma 2.4.

If no  $\psi_1$  occurs among the charge-2 slots, then the  $N - 1$  charge-2 slots are all equal to  $\psi_0^2$ , and the charge-3 slot is one of

$$\psi_2, \quad \psi_0^3, \quad \psi_0 \psi_1, \quad \psi_1 \psi_0.$$

This gives, respectively,

$$G_3, \quad G_4, \quad F_3, \quad F_4.$$

If one charge-2 slot is  $\psi_1$ , then the remaining charge-2 slots are copies of  $\psi_0^2$ , and the same four choices of the charge-3 slot give, respectively,

$$F_1, \quad F_2, \quad G_2, \quad G_1.$$

Thus the case with exactly one charge-1 slot gives precisely the eight displayed relations.

It remains to consider the case in which no slot has charge 1. Since the total charge is  $2N + 2$  and there are  $N + 1$  non-empty slots, every slot must have charge 2. Hence every slot is either  $\psi_0^2$  or  $\psi_1$ . If two slots are  $\psi_1$ , the identity vanishes by graded symmetry. If no slot is  $\psi_1$ , then the identity is

$$\Theta_N(\psi_0^2; (\psi_0^2)^N),$$

and every trace factor in its expansion is a trace of an even power of the odd word  $\psi_0$ ; hence it is zero.

The only remaining primitive identity is, up to graded symmetry,

$$\Theta_N(\psi_1; (\psi_0^2)^N).$$

By (3.16) and (3.23),

$$\Theta_N(\psi_1; (\psi_0^2)^N) = \text{tr}(\psi_0^{2N} \psi_1) = \frac{1}{2}(F_3 + F_4).$$

Thus the last remaining primitive identity also lies in the stated span.  $\square$

We now examine primitive identities of smaller charge. This is the only place where trace multiples of lower-charge identities can enter the charge- $2N + 2$  relation space.

**Lemma 3.4.** *There are no non-zero primitive identities of charge less than  $2N + 1$ . In charge  $2N + 1$ , up to scalar and graded symmetry, the only non-zero primitive identities are*

$$\Theta_N(\psi_0; (\psi_0^2)^N), \quad \Theta_N(\psi_0; (\psi_0^2)^{N-1}, \psi_1).$$

Moreover,

$$(3.24) \quad \text{tr}(\psi_0) \Theta_N(\psi_0; (\psi_0^2)^N) = -NG_4$$

and

$$(3.25) \quad \text{tr}(\psi_0) \Theta_N(\psi_0; (\psi_0^2)^{N-1}, \psi_1) = (N - 1)F_2 - \frac{1}{2}F_3 + \frac{1}{2}F_4.$$

*Proof.* A primitive identity has  $N + 1$  non-empty slots. If no slot has charge 1, then its charge is at least

$$2(N + 1) = 2N + 2.$$

If exactly one slot has charge 1, then its charge is at least

$$1 + 2N = 2N + 1.$$

If at least two slots have charge 1, then two slots are equal to the odd word  $\psi_0$ , and the identity vanishes by Lemma 2.4. Hence there are no non-zero primitive identities below charge  $2N + 1$ .

At charge  $2N + 1$ , there must be exactly one  $\psi_0$  slot and every remaining slot must have charge 2. The charge-2 words are  $\psi_0^2$  and  $\psi_1$ , and  $\psi_1$  can occur at most once. This gives exactly the two identities displayed in the statement.

By (3.17),

$$\Theta_N(\psi_0; (\psi_0^2)^N) = \text{tr}(\psi_0^{2N+1}).$$

Together with (3.22), this gives

$$\text{tr}(\psi_0)\Theta_N(\psi_0; (\psi_0^2)^N) = \text{tr}(\psi_0) \text{tr}(\psi_0^{2N+1}) = -NG_4,$$

which proves (3.24).

For the second product, multiply (3.18) by  $\text{tr}(\psi_0)$ . The term with  $a = 0$  in the sum vanishes because  $\text{tr}(\psi_0)^2 = 0$ . Hence

$$\begin{aligned} \text{tr}(\psi_0)\Theta_N(\psi_0; (\psi_0^2)^{N-1}, \psi_1) &= \text{tr}(\psi_0) \text{tr}(\psi_0^{2N-1}\psi_1) \\ &\quad - \frac{1}{N} \sum_{a=1}^{N-1} \text{tr}(\psi_0) \text{tr}(\psi_0^{2a+1}) \text{tr}(\psi_0^{2N-2a-2}\psi_1). \end{aligned}$$

On the other hand, using (3.21), (3.19) and (3.20), we compute

$$\begin{aligned} (N-1)F_2 - \frac{1}{2}F_3 + \frac{1}{2}F_4 &= \text{tr}(\psi_0) \text{tr}(\psi_0^{2N-1}\psi_1) \\ &\quad - \frac{1}{N} \sum_{a=1}^{N-1} \text{tr}(\psi_0) \text{tr}(\psi_0^{2a+1}) \text{tr}(\psi_0^{2N-2a-2}\psi_1). \end{aligned}$$

The two-trace terms

$$\text{tr}(\psi_0^{2a+1}) \text{tr}(\psi_0^{2N-2a-1}\psi_1)$$

cancel exactly between  $(N-1)F_2$  and  $-\frac{1}{2}F_3 + \frac{1}{2}F_4$ . This proves (3.25).  $\square$

**Proposition 3.5.** *For every  $N \geq 2$ ,*

$$\mathcal{K}_{N,2N+2} \subseteq \text{span}\{F_1, F_2, F_3, F_4, G_1, G_2, G_3, G_4\}.$$

*Proof.* By Lemma 3.1, every charge- $2N + 2$  relation is a linear combination of terms

$$M \cdot \Theta_N(W_0; W_1, \dots, W_N)$$

with all  $W_i$  non-empty and homogeneous.

If  $M = 1$ , then the primitive identity itself has charge  $2N + 2$ , and it lies in the required span by Lemma 3.3. If  $M > 0$ , then the primitive identity has charge strictly



less than  $2N + 2$ . By Lemma 3.4, the only possible non-zero primitive charge below  $2N + 2$  is  $2N + 1$ . Therefore  $M$  must have charge 1. The only charge-1 trace monomial is a scalar multiple of  $\text{tr}(\psi_0)$ , and the two possible products are precisely the products treated in (3.24) and (3.25). They also lie in the displayed span. This proves the proposition.  $\square$

#### 4. COEFFICIENT EXTRACTION AND INDEPENDENCE

It remains to prove that the eight spanning relations are independent. This will be done in the free signed trace algebra, before imposing any finite-rank trace identities.

Choose once and for all a canonical representative for every non-zero signed cyclic class of trace words. If a trace word is equal to its negative under a signed cyclic rotation, it is declared to be zero and no representative is chosen. We also choose a total order on the canonical trace words, and write products of trace factors in that order, using the supercommutation rule

$$\text{tr}(U) \text{tr}(V) = (-1)^{|U||V|} \text{tr}(V) \text{tr}(U).$$

If  $m$  is a canonical trace monomial and  $R \in \mathcal{T}$ , we write  $[m]R$  for the coefficient of  $m$  in  $R$ . A different choice of canonical representatives may change some rows below by non-zero signs, but it cannot affect the ranks of the coefficient matrices.

The free signed trace algebra is graded by letter content. In other words, one may count separately the number of occurrences of

$$\psi_0, \psi_1, \psi_2, \dots$$

appearing in a trace monomial. Signed cyclicity and supercommutation do not change this content. Hence trace polynomials of distinct letter content cannot cancel.

The eight relations split into the following five content blocks:

| relation        | letter content                |
|-----------------|-------------------------------|
| $F_1$           | $\psi_0^{2N-3} \psi_1 \psi_2$ |
| $F_2, F_3, F_4$ | $\psi_0^{2N} \psi_1$          |
| $G_1, G_2$      | $\psi_0^{2N-2} \psi_1 \psi_1$ |
| $G_3$           | $\psi_0^{2N-1} \psi_2$        |
| $G_4$           | $\psi_0^{2N+2}$               |

Here the entry in the second column records only content, not the cyclic order of a word. Thus the proof reduces to three one-dimensional blocks, one two-dimensional block, and one three-dimensional block.

**Lemma 4.1.** *The three one-dimensional blocks are non-zero. More precisely,*

$$(4.26) \quad [\text{tr}(\psi_0^{2N-3} \psi_1 \psi_2)] F_1 = \frac{1}{N},$$

$$(4.27) \quad [\text{tr}(\psi_0^{2N-1} \psi_2)] G_3 = 1,$$

$$(4.28) \quad [\text{tr}(\psi_0) \text{tr}(\psi_0^{2N+1})] G_4 = -\frac{1}{N}.$$

*Proof.* For  $F_1$ , the chosen monomial is a single trace. It can therefore only come from the tail term of  $F_1 = \Theta_N(\psi_0; (\psi_0^2)^{N-2}, \psi_1, \psi_2)$ . In the polarised tail, the trace word  $\text{tr}(\psi_0^{2N-3} \psi_1 \psi_2)$  appears exactly when the two odd defect slots  $\psi_1$  and  $\psi_2$  are adjacent in that cyclic order. There are  $(N-1)!$  labelled permutations with this property, and the polarising factor is  $1/N!$ . No odd inversion occurs between the two defect slots in these terms. Hence the coefficient is  $1/N$ .

For  $G_3$ , the chosen monomial again comes only from the tail term of  $G_3 = \Theta_N(\psi_0; (\psi_0^2)^{N-1}, \psi_2)$ . All  $N!$  labelled permutations give the same signed cyclic representative, and the polarising factor  $1/N!$  cancels them. This gives coefficient 1.

Finally, (4.28) is exactly the coefficient of  $\text{tr}(\psi_0) \text{tr}(\psi_0^{2N+1})$  in the collapse formula (3.22).  $\square$

**Lemma 4.2.** *With respect to the two coefficient functionals attached to*

$$m_1 = \text{tr}(\psi_0) \text{tr}(\psi_0^{2N-3} \psi_1) \text{tr}(\psi_1), \quad m_2 = \text{tr}(\psi_0) \text{tr}(\psi_0^{2N-3} \psi_1 \psi_1),$$

*and the column basis  $(G_1, G_2)$ , the coefficient matrix is*

$$(4.29) \quad \begin{array}{c|cc} & G_1 & G_2 \\ \hline m_1 & -\frac{1}{N(N-1)} & \frac{1}{N(N-1)} \\ m_2 & -\frac{1}{N(N-1)} & -\frac{1}{N(N-1)} \end{array}.$$

*Consequently  $G_1$  and  $G_2$  are linearly independent.*

*Proof.* The coefficient of  $m_1$  comes from the two-trace-block part of the polarised Cayley–Hamilton expression. One block is the single odd slot  $\psi_1$ . The other block contains all  $N-2$  copies of  $\psi_0^2$  and the even word

$$\psi_1 \psi_0 \quad \text{for } G_1, \quad \psi_0 \psi_1 \quad \text{for } G_2.$$

The remaining final trace is  $\text{tr}(\psi_0)$ . The labelled polarisation count is  $1/(N(N-1))$ . After moving the trace factors to canonical order, the two words  $\psi_1 \psi_0$  and  $\psi_0 \psi_1$  give opposite signed cyclic representatives. Thus

$$[m_1]G_1 = -\frac{1}{N(N-1)}, \quad [m_1]G_2 = \frac{1}{N(N-1)}.$$

The coefficient of  $m_2$  comes from the one-block trace part in which all inner slots lie in a single trace block, and the final trace is  $\text{tr}(\psi_0)$ . In this case the two cyclic placements have the same sign after normalisation, and the same polarisation count gives

$$[m_2]G_1 = [m_2]G_2 = -\frac{1}{N(N-1)}.$$

This proves (4.29). Its determinant is

$$\frac{2}{N^2(N-1)^2},$$

which is non-zero for  $N \geq 2$ .  $\square$

**Lemma 4.3.** *With respect to the three coefficient functionals attached to*

$$\begin{aligned} n_1 &= \text{tr}(\psi_0) \text{tr}(\psi_0^3) \text{tr}(\psi_0^{2N-4}\psi_1), \\ n_2 &= \text{tr}(\psi_0) \text{tr}(\psi_0^{2N-1}\psi_1), \\ n_3 &= \text{tr}(\psi_0^{2N}\psi_1), \end{aligned}$$

*and the column basis  $(F_2, F_3, F_4)$ , the coefficient matrix is*

$$(4.30) \quad \begin{array}{c|ccc} & F_2 & F_3 & F_4 \\ \hline n_1 & -\frac{1}{N(N-1)} & 0 & 0 \\ n_2 & \frac{1}{N} & -\frac{1}{N} & \frac{1}{N} \\ n_3 & 0 & 1 & 1. \end{array}$$

*For  $N = 2$ , the factor  $\psi_0^{2N-4}\psi_1$  in  $n_1$  is simply  $\psi_1$ . Consequently  $F_2, F_3, F_4$  are linearly independent.*

*Proof.* The coefficients are read from the explicit expansions (3.21), (3.19) and (3.20).

The monomial  $\text{tr}(\psi_0) \text{tr}(\psi_0^3) \text{tr}(\psi_0^{2N-4}\psi_1)$  occurs only in the three-trace sum of  $F_2$ , namely in the summand  $a = 1$  of the last sum in (3.21). Its coefficient is  $-1/(N(N-1))$ . Neither  $F_3$  nor  $F_4$  has a  $\psi_0^3$  slot, so  $n_1$  cannot occur in those two relations. This gives the first row.

The monomial  $n_2 = \text{tr}(\psi_0) \text{tr}(\psi_0^{2N-1}\psi_1)$  has coefficient  $1/N$  in  $F_2$ . In  $F_3$  and  $F_4$ , it is the  $a = 0$  term in the sums (3.19) and (3.20), and the signs are respectively  $-1/N$  and  $1/N$ . This gives the second row.

Finally,  $n_3 = \text{tr}(\psi_0^{2N}\psi_1)$  does not occur in  $F_2$ ; equivalently, the possible single-trace tail terms in  $F_2$  cancel because the two odd defect slots  $\psi_1$  and  $\psi_0^3$  are exchanged with the Koszul sign. In  $F_3$  and  $F_4$ , the monomial  $n_3$  is the tail term, with coefficient 1 in both cases. This gives the third row and proves (4.30). Its determinant is  $2/(N^2(N-1))$ , which is non-zero for  $N \geq 2$ .  $\square$

**Proposition 4.4.** *The eight relations*

$$F_1, F_2, F_3, F_4, G_1, G_2, G_3, G_4$$

*are linearly independent in the free signed trace algebra.*

*Proof.* The relations split by letter content into the five blocks displayed above. Trace polynomials of distinct letter content cannot cancel. The three one-dimensional blocks are non-zero by Lemma 4.1. The two-dimensional even block is independent by Lemma 4.2, and the three-dimensional odd block is independent by Lemma 4.3. Therefore the eight relations are linearly independent.  $\square$

**Remark 4.5.** The initial point  $N = 2$  is automatically covered by the preceding argument, but it is a useful check on the signs. The isolated coefficients become

$$[\text{tr}(\psi_0 \psi_1 \psi_2)]F_1 = \frac{1}{2}, \quad [\text{tr}(\psi_0^3 \psi_2)]G_3 = 1, \quad [\text{tr}(\psi_0) \text{tr}(\psi_0^5)]G_4 = -\frac{1}{2}.$$

The two non-trivial coefficient matrices reduce to

$$\begin{pmatrix} -\frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} -\frac{1}{2} & 0 & 0 \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 1 \end{pmatrix},$$

whose determinants are both  $1/2$ . Thus the base point of the diagonal already has four independent odd relations and four independent even relations. The general argument shows that inserting further copies of the even word  $\psi_0^2$  introduces no new phenomenon at this charge.

## 5. PROOF OF THE MAIN RESULT

*Proof of Theorem 1.* Each of the eight displayed elements is an instance of the signed polarised Cayley–Hamilton trace identity. Hence each lies in  $\mathcal{K}_{N,2N+2}$ , and therefore

$$\text{span}\{F_1, F_2, F_3, F_4, G_1, G_2, G_3, G_4\} \subseteq \mathcal{K}_{N,2N+2}.$$

The reverse inclusion is Proposition 3.5. Thus

$$\mathcal{K}_{N,2N+2} = \text{span}\{F_1, F_2, F_3, F_4, G_1, G_2, G_3, G_4\}.$$

By Proposition 4.4, these eight elements are linearly independent in the free signed trace algebra. Hence they form a basis of  $\mathcal{K}_{N,2N+2}$ . Their parities are immediate from their defining slots:

$$F_1, F_2, F_3, F_4 \in \mathcal{K}_{N,2N+2}^{\bar{1}}, \quad G_1, G_2, G_3, G_4 \in \mathcal{K}_{N,2N+2}^{\bar{0}}.$$

Consequently

$$\dim \mathcal{K}_{N,2N+2}^{\bar{1}} = 4, \quad \dim \mathcal{K}_{N,2N+2}^{\bar{0}} = 4.$$

This proves the theorem.  $\square$

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